

Experimental and Numerical Simulation Analyses of the Interaction between Fine-grained Sediments and Low-velocity Water Flow: A Case Study of the Alar Section in the Upper Tarim River

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ABSTRACT

This study examined the water–sediment feedback mechanisms of fine sediment (0.1–0.3 mm) in the upper reaches of the Tarim River under low-flow velocities (0.12–0.32 m/s) through a combination of flume experiments and Eulerian two-phase flow simulations in Fluent. The standard k-ε turbulence model was employed to systematically evaluate nine conditions representing three sediment sizes and three flow velocities. The results show that (1) when the particle size is less than 0.2 mm and flow velocity is 0.12 m/s, the sediment remains under uniform settling condition; when the particle size exceeds 0.2 mm, deposition accumulates; and when the flow velocity exceeds 0.12 m/s, deposit thickness decreases. (2) The flow velocity primarily controls sediment transport intensity, whereas particle size ≥ 0.2 mm significantly alters the turbulent structure of the flow. (3) When the flow velocity is ≥ 0.22 m/s and particle size is ≥ 0.2 mm, turbulent kinetic energy in the sediment settling zone increases by an additional 7.23%, indicating a synergistic enhancement effect.

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1. INTRODUCTION

Sustainable utilization of Tarim River's water resources is crucial for regional ecological balance and economic development. However, the deposition behaviour of fine sediment (particle size: 0.1–0.3 mm) in this river differs significantly from coarse sediment owing to its lower settling velocity and greater tendency to remain suspended (Deng 2014). This results in a complex and unpredictable sediment transport process. Especially during the dry season (when flow velocity is typically low, ranging from 0.12 m/s to 0.32 m/s), sediment deposition intensifies, directly constraining the water conveyance capacity of the river channel and threatening the health of the ecosystem (Fig. 1).

Vegetation plays a key role in regulating water–sediment processes and has therefore attracted considerable research attention. Previous studies have demonstrated that vegetation type (floating/rigid) and density can significantly alter the spatial distribution of sediment deposition, for example, disrupting streaks of sediment deposition, inducing above-threshold

sedimentation, and increasing sediment flux, by modifying flow structures and dissipating turbulent kinetic energy (TKE) (Gui et al. 2024; Sun et al. 2023; Han et al. 2023; Yang et al. 2022; Soler et al. 2021; Li et al. 2018). These results consistently indicate that vegetation type and density are the key regulatory factors shaping the spatial pattern of sediment deposits within the river channel.

Sediment deposition is inherently a complex phenomenon that involves the coupling of multiple physical fields (flow and sediment fields) and coordinated control by multiple factors. Numerous studies have revealed the significant effects of hydraulic characteristics (e.g. flow velocity, turbulence intensity, blockage effect, wake structure) (Kim et al. 2018; Hu et al. 2018), sediment properties and river engineering interventions (e.g. sediment retention in dams, reservoir sediment recharge strategies, and sediment erosion mitigation measures) (Huang et al. 2025; Fu et al. 2023; Brenna et al. 2025; Daneshfaraz et al. 2026), and advanced monitoring and simulation methods (e.g. particle size distribution correction, acoustic signal recognition, intelligent algorithm models, gravity current experiments, initial flow velocity



Fig. 1 Sediment deposition morphology of the Tarim River Basin:(a) canal sedimentation, (b) irrigation channel siltation, (c) riverbank siltation, (d) silt thickness.

threshold models, ultrasound-assisted flocculation) (Jing et al. 2024; Wu et al. 2023; Li et al. 2024; Cheng et al. 2020; Zordan et al. 2018) on sediment transport and deposition.

Numerical simulation serves as a core tool for analysing complex sediment processes, and significant progress has been achieved in revealing excess pore pressure dissipation (Sun et al. 2019), characterizing deposition morphology in special geomorphologies (Zhang et al. 2017), demonstrating sediment reduction effects via structural flow disturbance (Zhang et al. 2023), developing advanced algorithms such as the splitting operator method (Mrad et al. 2022), predicting sediment accumulation locations (Lakzian et al. 2020), simulating gravity-induced slip (Jia et al. 2013), and understanding slope-velocity equilibrium mechanisms (Janocko et al. 2013).

In addition, the properties of fine-grained sediments are highly complex, especially when interacting with water flow. Generally, fine-grained sediments comprise inorganic sand, silt, and clay particles with diameters less than 2 mm, and their presence can significantly alter river hydraulics and geomorphic features (Pietrzyk et al. 2022). When fine-grained sediments accumulate in lowland rivers, they may negatively impact water quality and riverbed habitats, underscoring the importance of understanding their dynamic processes (Chuenchum et al. 2023). In such systems, cohesive sediments, such as silt and clay, exhibit distinct erosion and deposition thresholds because of electrochemical effects, leading to flocculation and hysteresis during changes in flow conditions (Stone et al. 2021).

Although the above-mentioned studies have greatly enhanced our understanding of sediment dynamics, existing research on vegetation-water-sediment interaction and most models and experimental studies mainly focused on non-desert environments or steady flow conditions. There is still a lack of research on the coupling and synergistic effects of flow dynamics and sediment characteristics of fine-grained silt (0.1–0.3 mm) in the Tarim River under low flow-velocity conditions. This directly leads to inaccurate prediction of sediment deposition dynamics in the upper reaches of the Tarim River.

Therefore, the focus of this study was on the Alar section of the upper reaches of the Tarim River. Fine-grained sediment samples and flow data were collected on-site. The Fluent software system was used to simulate the sedimentation process of fine sand of different particle sizes under low flow conditions. The water-sediment

coupling effect was analysed in detail, and the numerical simulation results were verified. The key feedback mechanism between fine-grained sediment and low flow velocity in the upper reaches of the Tarim River was revealed.

2. RESEARCH METHODOLOGY

The aim of this study was to investigate the physical mechanisms governing the interaction between fine sediment and low-velocity water flow. Given that the Alar reach in the upper Tarim River is a typical braided channel characterized by highly dynamic bed morphology and extremely sparse vegetation cover, the roles of riverbed and vegetation were deliberately excluded from the current scope. The primary focus was on quantifying the bimodal coupling effects between hydrodynamic forces and inherent sediment properties (particle size). Because accurately measuring in-channel sediment velocities is difficult, a methodology integrating numerical simulation with experimental validation was used.

2.1 Sediment Sampling, Measurement, and Analysis

Sediment samples were collected at the beginning of siltation (sampling site 1), end of siltation (sampling site 2), the river bank (sampling site 3), and the centre of sediment deposition (sampling site 4) in a segment of the upper reaches of the Tarim River in Alar. The river flow velocities measured at the sediment sampling sites are shown in Fig. 2.

Experimental analysis revealed that fine sediment particles with a diameter ranging from 0.075 mm to 0.35 mm collected from a segment of the Tarim River in Alar accounted for more than 95% of all particles. The rest of the particles, which accounted for 5%, comprised fine soil. The flow velocity of the Tarim River in the dry season (October to February of the following year) ranged from approximately 0.1 m/s to 0.3 m/s.

The mutual feedback mechanisms of fine sediment (particle sizes: 0.1 mm, 0.2 mm, 0.3 mm) and low-velocity water flow (flow velocities: 0.12 m/s, 0.22 m/s, 0.32 m/s) were analysed and verified using flume experiments.

2.2 Experimental Protocol

The experiments were conducted in a rectangular glass flume with a length of 12 m, a width of 0.6 m, and a height of 0.8 m. The flow rate was regulated by a variable-frequency pump and an adjustable inlet valve, corresponding to a mainstream flow velocity ranging from 0.12 to 0.32 m/s. An adjustable tailgate was installed at the

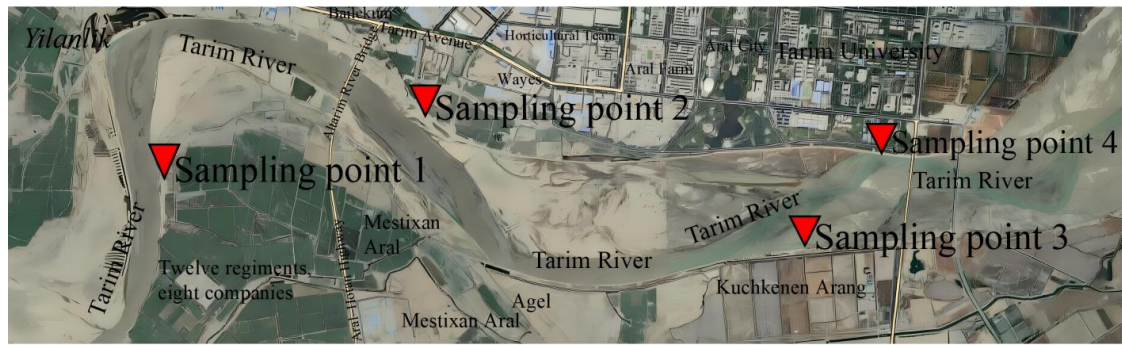


Fig. 2 Locations of the sampling sites

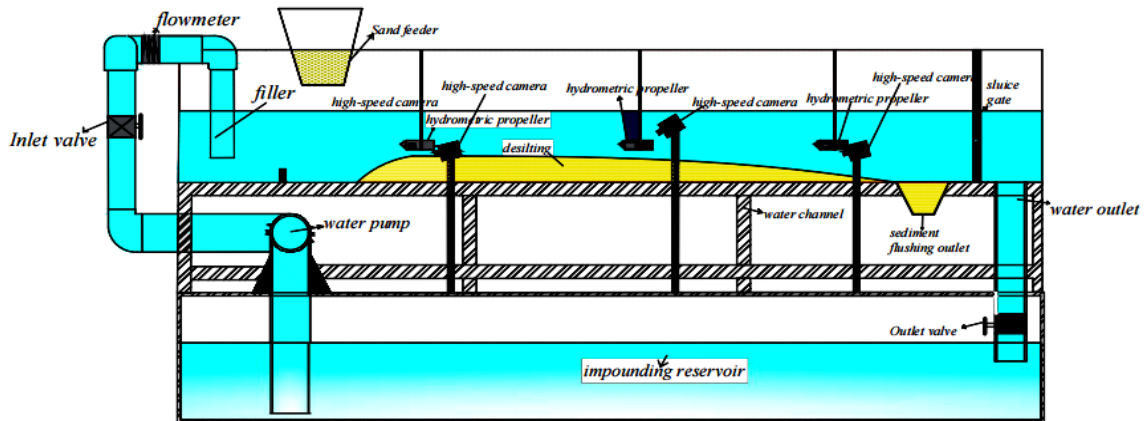


Fig. 3(a) Schematic of the test flume device



Fig. 3(b) Photograph of the test flume device

downstream end of the flume to maintain a constant water depth under various test conditions. A plan and profile view of the flume arrangement is shown in Fig. 3b (Dharmadhikari & Gandhi, 2025; Gandhi, 2024).

Sediment was fed through a quantitative sediment discharge port located at the upstream end of the flume, directly above the central axis. A cyclone mixer ensured uniform delivery of the sediment–water mixture. Downstream of the deposition zone, a multi-stage sand discharge gate and a sedimentation pond were installed to separate suspended load from bed load in real time, allowing the recirculating water–sediment system to operate under dynamic equilibrium.

Experimental procedure. Before each test, the pump and tailgate were adjusted to achieve the target flow velocity (0.12, 0.22, or 0.32 m/s) and a constant water depth. After the flow stabilized for 5 minutes, the background flow field was recorded. Subsequently, pre-weighed fine sediment (particle sizes of 0.1, 0.2, and 0.3 mm, respectively) was introduced through the sediment discharge port at a constant feeding rate. From the moment

sediment addition began, the deposition process was synchronously observed and recorded using high-speed cameras and a velocimeter. The velocimeter was positioned at three characteristic cross-sections (0.2 m, 0.5 m, and 1.0 m downstream of the sediment inlet). Each experimental condition was repeated three times to ensure reproducibility. After completing a set of tests, all residual sediment was carefully removed from the flume, and the system was flushed with clean water before proceeding to the next set of tests (Dharmadhikari & Gandhi, 2025; Gandhi, 2024).

Based on predetermined test conditions, the flow conditions were controlled through the hydraulic control device, and sediment input regimes were simulated using the sediment delivery device. During the tests, data on vertical flow velocity profiles and sediment deposition at characteristic cross-sections (0.2 m, 0.5 m, and 1.0 m) of the flume were collected in real-time by using Doppler velocimeters and high-speed cameras. This enabled a more intuitive analysis of sediment deposition morphology and variation patterns. The actual test flume device is shown in Fig. 3b.

2.3 Numerical Calculation Methods

To elucidate the mutual feedback mechanisms of water and sediment, particularly the movement of fine sediment under low-velocity conditions, it is essential to investigate three-dimensional vertical flow velocities and their interactions with sediment dynamics. Accordingly, we constructed a three-dimensional model based on the aforementioned experimental flume. Guided by the numerical simulation theory for water–sediment two-phase flow, this model simulates the coupled dynamics between water and sediment within the flume under varying sediment particle sizes (using field-collected samples) and hydraulic conditions.

2.3.1 Numerical simulation theory of water–sediment two-phase flow

In this study, the deposition characteristics of fine sediment (0.1–0.3 mm) under low-velocity conditions (0.12–0.32 m/s) in the upper Tarim River were addressed using Eulerian two-fluid model to describe the water–sediment two-phase flow. The sediment phase was treated as a continuous fluid medium that interpenetrates and interacts in a coupled manner with the water phase. The model ensures conservation of mass and momentum through the following governing equations:

1. Continuity Equation

The continuity equations for the water and sediment phases are as follows:

$$\frac{\partial(\alpha_l)}{\partial t} + \nabla \cdot (\alpha_l \vec{u}_l) = 0 \quad (1)$$

$$\frac{\partial(\alpha_s)}{\partial t} + \nabla \cdot (\alpha_s \vec{u}_s) = 0 \quad (2)$$

where ρ_l denotes water density, α_l denotes the volume fraction of the water phase, $\vec{u}_l$ denotes the velocity vector of the water phase, t denotes time, ∇ denotes the Hamiltonian operator, ρ_s denotes sediment density, α_s denotes the volume fraction of the sediment phase, $\vec{u}_s$ denotes the velocity vector of the sediment phase, and S_s denotes the source term of the sediment phase.

Applicability Notes: Equation (1) conserves water mass and applies to incompressible flow, as the experimental flow Reynolds number ($Re < 5000$) falls within the laminar-weak turbulent transition regime. Equation (2) incorporates the source term S_s to represent sediment settling/suspension processes, specifically addressing the high suspendibility of fine sediment (0.1–0.3 mm) under low-velocity conditions. The settling velocity in this term is calculated using modified Stokes' law (considering particle Reynolds number $Rep < 1$).

2. Momentum Conservation Equations

The momentum conservation equations for the water and sediment phases are as follows:

$$\frac{\partial(\alpha_l)}{\partial t} + \nabla \cdot (\alpha_l \vec{u}_l) = -\alpha_l \nabla p + \nabla \cdot \vec{\tau}_l + \alpha_l \vec{g} + \sum_{k \neq l} \vec{F}_{lk} \quad (3)$$

$$\frac{\partial(\alpha_s)}{\partial t} + \nabla \cdot (\alpha_s \vec{u}_s) = -\alpha_s \nabla p + \nabla \cdot \vec{\tau}_s + \alpha_s \vec{g} + \sum_{k \neq s} \vec{F}_{sk} \quad (4)$$

where p denotes pressure, $\vec{\tau}_l$ denotes the viscous stress tensor of the water phase, $\vec{g}$ denotes the gravitational acceleration vector, $\vec{F}_{lk}$ denotes the interaction force between the water and sediment phases, and $\vec{\tau}_s$ denotes the viscous stress tensor of the sediment phase.

For the coupled flow mechanisms of fine sediment particles under low-flow conditions, the Schiller-Naumann drag model was used. This model demonstrates good insensitivity to variations in the drag coefficient due to the dominance of viscous forces and limited ability of particles to modulate turbulence when $Rep < 800$ (in this study, $Rep < 50$). Therefore, the model provides reliable predictions for simulating the behaviour of fine particles in low-speed flows (Zou et al. 2008; Pietrzyk et al. 2022; Bass & Worth Longest 2018).

3. Selection and Parametric Adaptability of the Turbulence Model

For the rectangular channel and low-velocity conditions in the flume experiments, the standard k - ϵ model was used to characterize the turbulence effects:

$$\frac{\partial(\rho k)}{\partial t} + \nabla \cdot (\rho u k) = \nabla \cdot \left[\left(u + \frac{\mu}{\sigma_k} \right) \nabla k \right] + G_k - \rho \epsilon \quad (5)$$

$$\frac{\partial(\rho \epsilon)}{\partial t} + \nabla \cdot (\rho u \epsilon) = \nabla \cdot \left[\left(u + \frac{\mu}{\sigma_\epsilon} \right) \nabla \epsilon \right] + C_{1\epsilon} \frac{\epsilon}{k} G_k - C_{2\epsilon} \rho \frac{\epsilon^2}{k} \quad (6)$$

where k denotes turbulence kinetic energy, ρ denotes mixture density, u denotes the molecular viscosity coefficient, u_t denotes the turbulent viscosity coefficient, and σ_k is the turbulent Prandtl number of the turbulent kinetic energy k (it is set at 1.0 to enhance the sensitivity of the low-flow velocity sedimentation prediction), G_k denotes the turbulence kinetic energy generation term due to the average velocity gradient, ϵ denotes the turbulent dissipation rate, and $C_{1\epsilon}$ and $C_{2\epsilon}$ are empirical parameters set to 1.44 and 1.92, respectively, according to the Launder standard values (Elghobashi 2019).

Applicability Note: The k - ϵ standard model demonstrated robust performance in low-velocity open-channel flows, with its isotropic turbulence assumption being well-suited to the weak secondary flow conditions encountered in our experiments (Rodi 2017).

2.3.2 Numerical model construction and meshing

Based on the flume experiments, the region with significant sediment deposition was identified within 3 m downstream of the sediment inlet. Given the substantial impact of the terminal sedimentation pond on water–sediment simulation results, a 3D model was constructed using Creo software with the actual dimensions of the deposition zone (length $\times$ width $\times$ height = 3 m $\times$ 0.3 m $\times$ 0.8 m) to ensure numerical reliability. The constructed model was imported into Fluent for meshing. Because of structural simplicity, a structured meshing approach was adopted. Subsequent grid refinement was applied to the flume boundary regions, areas with large sediment-concentration changes, and critical zones exhibiting large gradients in flow velocity and sediment concentration. A coarser mesh was used in regions exhibiting relatively small spatial variations, as illustrated in Fig. 4.

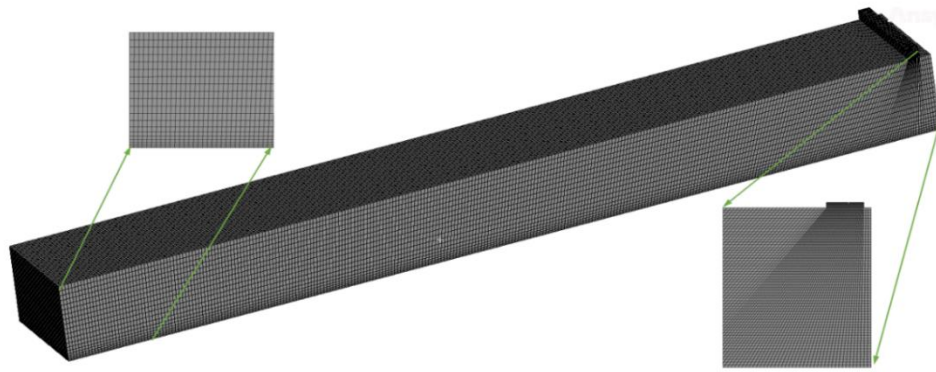


Fig. 4 Meshed flume model

2.3.3 Computational assumptions and boundary conditions

Given the depositional characteristics of fine sediment and hydrodynamic features in channel flows, the following assumptions were adopted for fine sediment (0.1–0.3 mm) motion under low-velocity conditions (0.12–0.32 m/s): (1) No interphase exchange: negligible mass/energy transfer between phases and neglect of inter-particle collisions. (2) Continuous sediment phase: sediment phase tracked as an incompressible continuous fluid medium with constant density. (3) Rigid spherical particles: fine particles treated as rigid spheres, while ignoring the electrochemical flocculation effects of natural sediments; salinity measured in the Alar section was less than 0.5 g/L, making the rigid-sphere assumption acceptable (Gupta et al. 2023). (4) Incompressible isothermal flow: phase change excluded; fluid density invariant to temperature changes. (5) Simplified boundary: hydraulically smooth PVC bed (roughness height = 0.01 mm), with negligible boundary layer and vortex effects.

Notably, when the sediment volume fraction near the bed exceeded 0.25, inter-particle collisions were theoretically non-negligible. However, under the low-velocity conditions (0.12–0.32 m/s) of this study, the particle Stokes number was < 0.1 , and settling dominated over collisional effects. A future study needs to incorporate the kinetic theory of granular flow for further refinement.

Velocity inlet is used to define the inlet conditions of the water and sediment phases. The water flow velocity measured on-site was used as the initial velocity of the water flow (main phase). The sediment particles of the Tarim River are very small, so the flow has a strong sediment entrainment capacity. The initial velocity of the sediment (secondary phase) at the inlet was assumed to be the same as the initial water flow velocity, and the

sediment content was measured through on-site sampling. The Granular option was selected to set the particle diameter, density, and viscosity models, and the turbulence parameters were specified using the standard $k-\varepsilon$ model.

At the outlet, the sediment particles can be carried out of the simulated area by the flow. Because the outflow is natural, the free outflow boundary condition was used. The roughness of the flume wall has a great influence on flow resistance and overall hydrodynamic conditions. The bed roughness of the Tarim River is relatively small because of its relatively flat and sandy riverbed, so the boundary condition of the flume wall was set as that of a non-slip wall. The Schiller-Naumann model of the drag coefficient was defined in the Phase Interaction dialog box of Fluent.

2.3.4 Mesh independence verification

Table 1 summarizes the maximum bottom pressure and TKE under six grid densities (0.29 million, 1.1 million, 1.9 million, 2.19 million, 2.88 million, and 3.36 million elements). When the number of grid elements was ≤ 1.1 million, significant differences in TKE distribution along the Y-direction were observed (maximum deviation $> 15\%$), indicating that the solution has not yet converged. As the grid increased to 1.9–2.19 million elements, the pressure distribution stabilized (deviation $< 3\%$), but TKE fluctuations remained above 5%. When the grid reached ≥ 2.88 million elements, both the pressure and TKE curves converged (pressure change rate, 0%, TKE change rate, -0.25%), satisfying the grid independence requirements for both physical fields simultaneously.

According to the ASME V&V 20-2009 standard mandate for grid convergence studies to include at least four distinct mesh densities (Roache et al. 2017), in this study, seven mesh sets ranging from 0.9 to 3.36 million

Table 1 Grid independence verification: Convergence of key parameters with increasing grid number

Grid count (million)	Maximum bottom pressure (Pa)	Change rate (%)	Inlet TKE (m^2/s^2)	Change rate (%)
0.90	1.05	-	0.0239	-
1.11	134.76	+12734	0.3106	+1199
1.91	191.61	+42.2	0.5124	+65.0
2.19	238.98	+24.7	0.6379	+24.5
2.88	257.35	+7.7	0.7997	+25.4
3.36	257.35	0.0	0.7977	-0.25

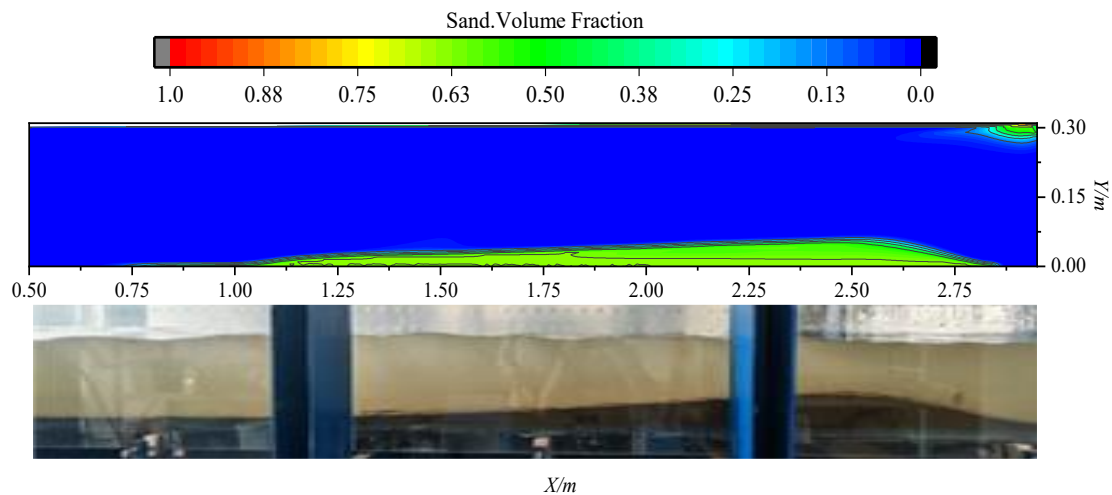


Fig. 5 Numerically simulated and experimentally observed water and sediment flow regimes

were used. The calculated convergence ratio ($RG = 0.98$, where the ideal value ≈ 1) confirmed that the 2.88-million mesh was within the asymptotic convergence zone. Consequently, the 2.88-million mesh configuration was selected for subsequent numerical simulations.

3. COMPARISON AND DEMONSTRATION OF THE EXPERIMENTAL AND NUMERICALLY SIMULATED RESULTS

To verify the model’s applicability for analysing the sedimentation characteristics of fine sediment, experimental validation of the numerical method’s accuracy was conducted. Based on similarity principles, geometric similarity was achieved as described in Section 2.3.2, whereas dynamic similarity was validated through direct comparison of sediment volume fraction (C_v) between the numerical and experimental results. C_v , defined as the volumetric proportion of sediment particles per control unit, serves as the key governing variable that determines the interphase forces and sediment settling flux. Velocity comparisons further verified kinematic similarity.

As shown in Fig. 5, C_v distributions from the numerical simulation and experiments revealed high consistency. Both approaches captured significant bed enrichment of fine sediment in the near-bed high- C_v zone ($C_v > 0.25$, $Y < 0.1$ m) with less than 8% error, confirming

gravity-dominated sedimentation mechanisms. This comparison validated the model’s reliability in predicting the critical parameter C_v and established a solid foundation for subsequent analysis of mutual feedback mechanisms.

To validate the accuracy of the numerical simulation, the simulated flow velocities were compared with experimental measurements for three different particle sizes (0.1 mm, 0.2 mm, and 0.3 mm). Quantitative assessments were conducted by calculating the root mean square error (RMSE) and mean absolute percentage error (MAPE). The results are summarized in Table 2. For the 0.1 mm particle size, the RMSE was 0.0318 m/s and MAPE was 18.9%; for 0.2 mm, RMSE was 0.0155 m/s and MAPE was 9.3%; and for 0.3 mm, RMSE was 0.0201 m/s and MAPE was 15.0%. These statistical indicators demonstrate good agreement between the numerical simulations and experimental data, especially for the 0.2 mm particle size. The error for the 0.1 mm particle size was relatively larger (maximum absolute error, 0.054 m/s), but still within the acceptable range (Daneshfaraz et al. 2025; Yilmaz et al. 2026).

The comparative results prove the effectiveness of the standard $k-\epsilon$ model in analysing the sedimentation and siltation characteristics of fine sediment, and the follow-up research presented in this paper is also based on this model.

Table 2 Error analysis of the comparison between numerical simulation and experimental flow velocity

Sediment Grain size	Section position	Simulated velocity (m/s)	Experimental velocity (m/s)	Absolute error (m/s)	RMSE (m/s)	MAPE (%)
0.1	0.2	0.171	0.117	0.054	0.0318	18.9
0.1	0.5	0.143	0.152	-0.009		
0.1	1.0	0.132	0.126	0.006		
0.2	0.2	0.121	0.118	0.003	0.0155	9.3
0.2	0.5	0.122	0.137	-0.015		
0.2	1.0	0.132	0.154	-0.022		
0.3	0.2	0.154	0.134	0.020	0.0201	15.0
0.3	0.5	0.163	0.141	0.022		
0.3	1.0	0.142	0.124	0.018		

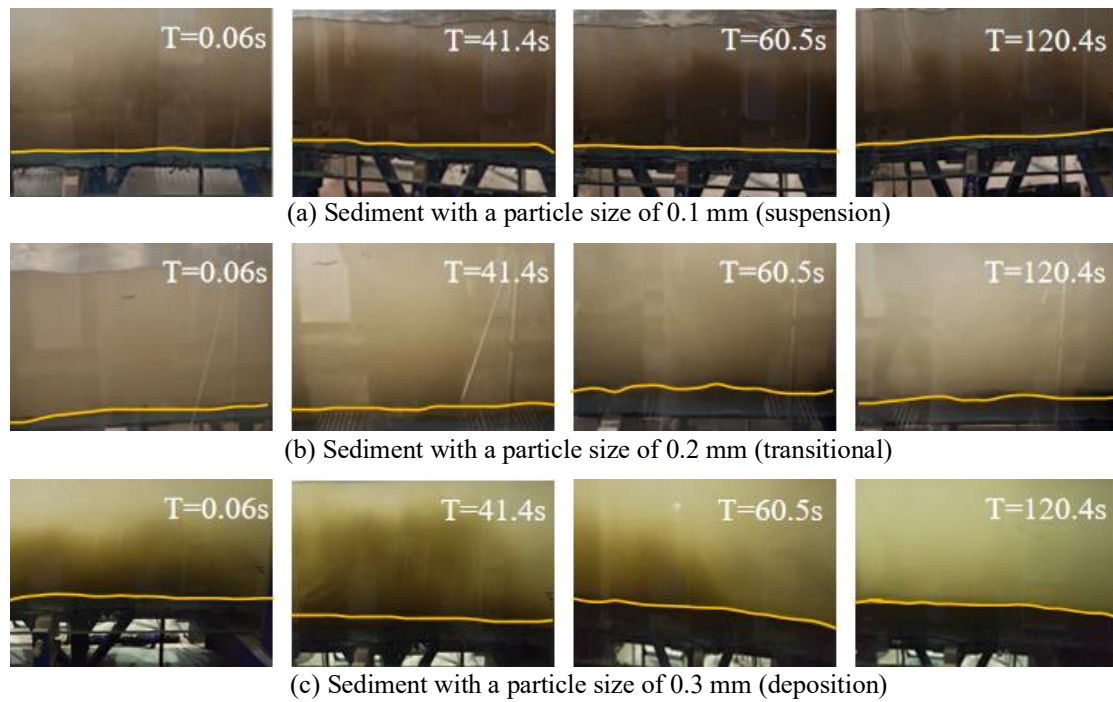


Fig. 6 Coupling of water–sediment flow regimes for sediment with different particle sizes

4. RESULTS

4.1 Mutual Feedback Mechanism of Flow Regimes of Water and Fine-Grained Sediment

The flow regimes of water and sediment directly reflect the dynamic coupling between flow and fine sediment. By observing water–sediment coupling dynamics at different time scales ($T = 0.06\text{--}120.4\text{ s}$) in the experimental flume, the influence of sediment particle size ($0.1\text{--}0.3\text{ mm}$) on water–sediment distribution patterns was elucidated. Numerical simulations further generated C_v (sediment volume fraction) contour plots under varying flow velocities and particle sizes to delineate sediment transport regimes.

As shown in Fig. 6a, fine sediment (0.1 mm) rapidly achieved uniform suspension within the water column, forming a diffuse distribution. Fine (0.1 mm) sediment was quickly and uniformly suspended in the water column to form a diffuse distribution (Fig. 6a). With the passage of time ($T = 41.4\text{--}120.4\text{ s}$), the sediment was suspended (Fig. 6a–c). Fine (0.2 mm) sediment showed some suspension characteristics at the initial stage ($T = 0.06\text{ s}$), but with the prolongation of flow time ($T = 41.4\text{ s}$), the effect of gravity gradually appeared (Fig. 6b). Fine (0.3 mm) sediment showed a significant deposition trend at $T = 0.06\text{ s}$, and accretion of fine sediment was noticeable with the extension of the flow time (Fig. 6c). The analysis showed that fine sand with a particle size of 0.1 mm remained suspended over time, whereas fine sand with a particle size ranging from 0.2 mm to 0.3 mm exhibited alternating phases of deposition and resuspension.

Furthermore, the relationships between fine sediment deposition and flow velocity and between fine sediment deposition and particle size were analysed using numerical simulations. Fine sediment (0.1 mm) exhibited a quasi-uniform suspension state in the flow because of its

small settling rate (Fig. 7a). As the particle size increased to 0.2 mm , a significant vertical gradient developed under the effect of gravity (Fig. 7b). The fine sediment (0.3 mm) settled quickly, and a well-defined water–sediment interface appeared 0.70 m below the water surface (Fig. 7c).

The analysis of flow velocity ($0.12\text{--}0.32\text{ m/s}$) showed that, at a flow velocity of 0.12 m/s , sediment volume fraction (C_v) exhibited a relatively uniform vertical distribution (Fig. 8a). This indicates the gravity settlement and sediment entrainment capacity of the flow are nearly balanced. When the flow velocity increased to 0.22 m/s , the sediment entrainment capacity of the flow increased, the peak C_v value at the bottom migrated to the middle reaches (migration distance $\Delta L = 0.5\text{ m}$), and the vertical gradient increased significantly. When the flow velocity reached 0.32 m/s , the peak value of C_v at the bottom continued to migrate (migration distance $\Delta L = 0.5\text{ m}$), and the spatial heterogeneity of C_v increased. This result is consistent with the flow velocity–deposition positive feedback mechanism proposed by [Hu et al. \(2018\)](#)**Error! Reference source not found..**

Overall, when the sediment particle size is less than 0.2 mm and the flow velocity is 0.12 m/s , fine sand will maintain uniform flow and settle. When the sediment particle size is greater than 0.2 mm , accumulation will be observed. When the flow velocity is greater than 0.12 m/s , the sediment deposition thickness will decrease accordingly.

4.2. Mutual Feedback Mechanism of Fine Sediment and Flow Velocity

The vertical velocity component and its induced turbulence intensity are particularly sensitive to the suspension, deposition, and transport processes of fine sediment, constituting a key to understanding water-sediment

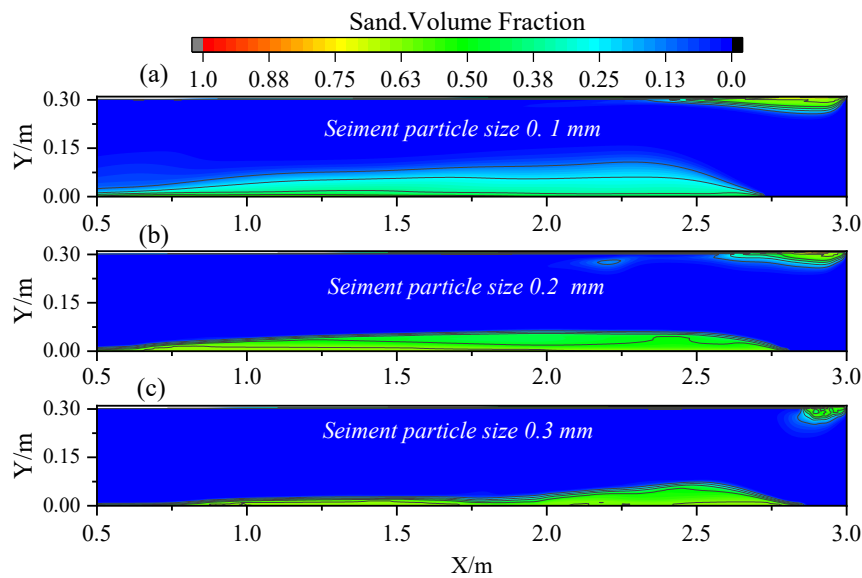


Fig. 7 Vertical distribution of sediment with different particle sizes

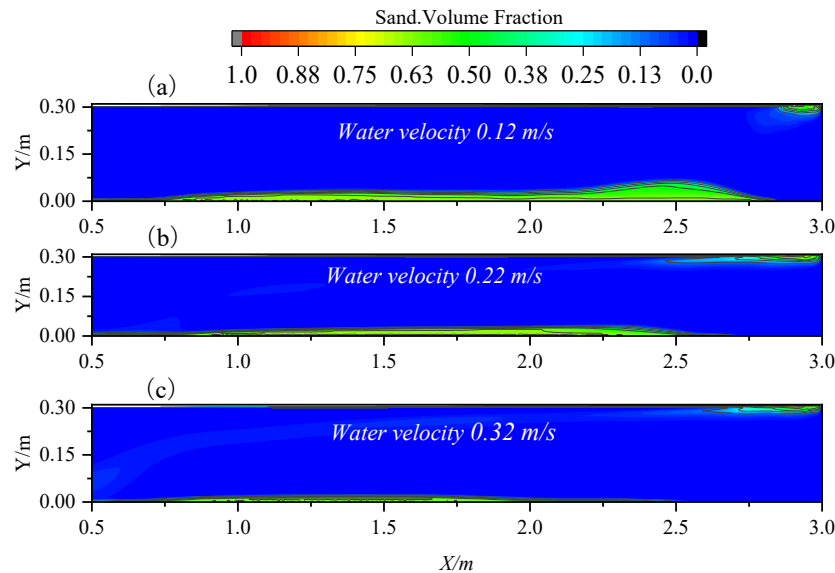


Fig. 8 Vertical distributions of C_v in sediment deposition at different flow velocities

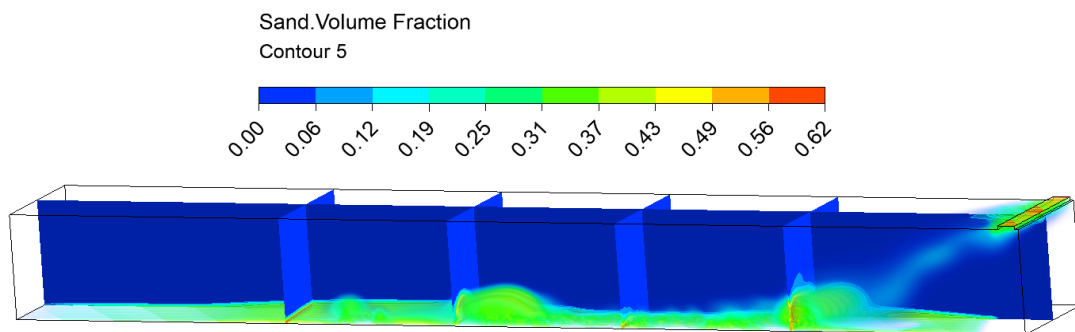


Fig. 9 Three-dimensional flow velocity monitoring diagram of a typical section

coupling. Therefore, through three-dimensional flow-velocity monitoring at characteristic channel cross-sections (Fig. 9), the governing patterns of flow velocity (0.12–0.32 m/s), sediment particle size (0.1–0.3 mm), and their coupling effects on the vertical distributions of both flow and sediment velocities were systematically analysed in this study (Figs. 10 and 11).

1. Impact of Flow Velocity on Vertical Sediment Velocity Distribution:

The impact of flow velocity on the vertical distribution of sediment velocity was investigated through simulations of sediment velocity profiles at four characteristic locations ($X = 0.2$ m, 0.5 m, 1.0 m, and 1.8 m; vertical range $Z = 0.13$ – 0.27 m) under three inlet flow

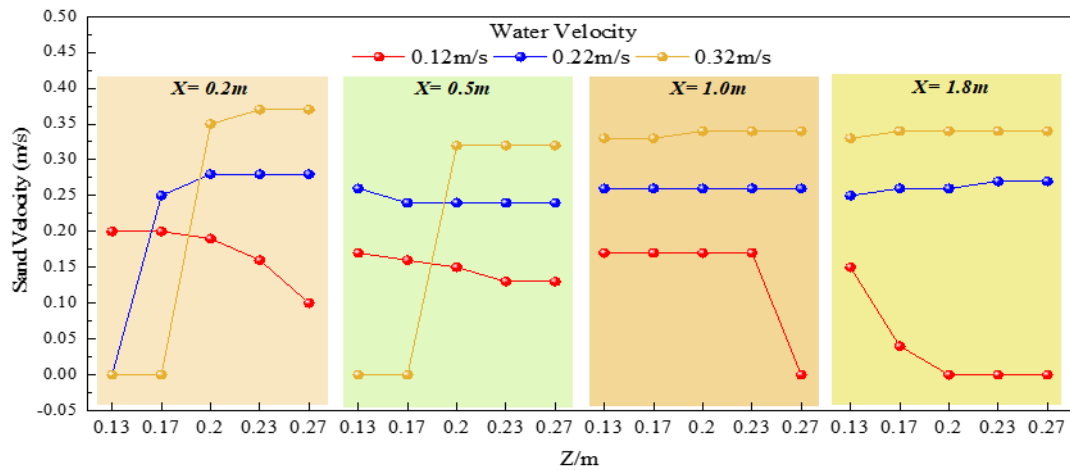


Fig. 10 Curves of sediment flow velocities at different water flow velocities

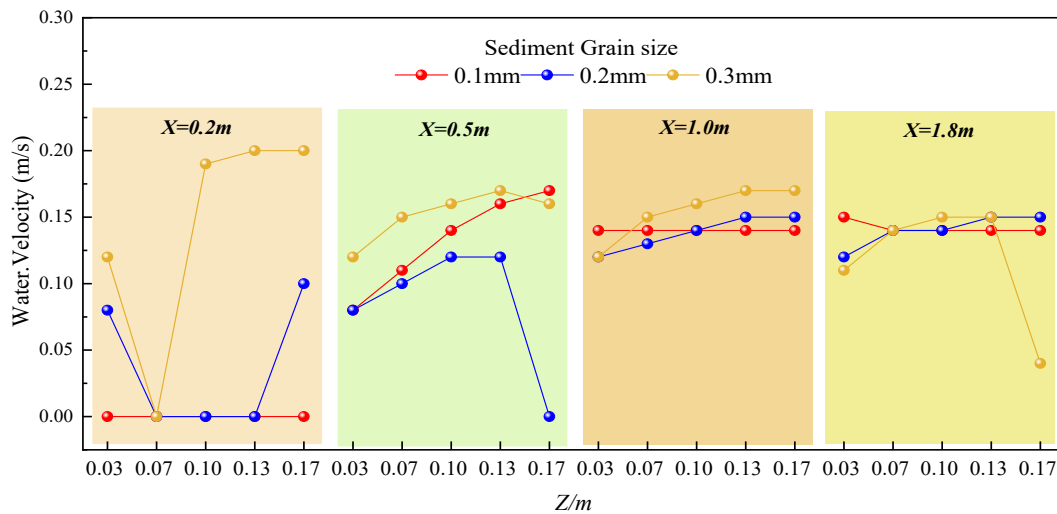


Fig. 11 Water flow velocities at different profiles for different sediment particle sizes

velocities (0.12 m/s, 0.22 m/s, and 0.32 m/s), as presented in Fig. 10. The sediment velocity increased significantly as the water flow velocity increased. Specifically, the maximum sediment velocity corresponding to a flow velocity of 0.32 m/s was 0.35 m/s ($X = 0.2$ m), which is 40% higher than that corresponding to a flow velocity of 0.12 m/s. In terms of spatial distribution, sediment velocity distribution in the near-inlet segment ($X \geq 1.0$ m) tended to be uniform (fluctuation amplitude < 0.05 m/s), whereas the sediment velocity in the outlet segment ($X = 0.2$ m) fluctuated drastically (e.g. the fluctuation amplitude reached 0.10 m/s at a flow velocity of 0.22 m/s). Sediment movement in the inlet section was mainly dominated by flow velocity, presenting relatively stable transport characteristics; in the outlet section, the sediment settled down and vertical turbulence significantly intensified, leading to an increase in the uncertainty of sediment movement.

2. Impact of Sediment Particle Size on Vertical Flow Velocity Distribution:

Vertical flow velocity profiles at four characteristic cross-sections were simulated ($X = 1.8$ m [inlet segment], 1.0 m, 0.5 m, and 0.2 m [outlet segment]; $Z = 0.03$ –0.17 m) under three sediment particle sizes (0.1 mm, 0.2 mm, and 0.3 mm), as shown in Fig. 11. The experimental

results reveal significant particle size effects and longitudinal evolution patterns.

(1) Particle size effect: At a particle size of 0.3 mm, the maximum vertical flow velocity reached approximately 0.20 m/s with vertical velocity fluctuation amplitude increasing to 0.11 m/s, indicating coarser sediment significantly enhances vertical mixing intensity and turbulence.

(2) Longitudinal evolution: Flow velocity distributions remained relatively uniform in the inlet segment ($X = 1.8$ m; fluctuation < 0.05 m/s for all sizes), whereas vertical velocity gradients and fluctuation amplitudes intensified markedly toward the outlet segment ($X = 0.2$ m).

(3) Critical particle size characterization: Flow intensity increased substantially when the particle size ≥ 0.2 mm. These findings directly confirmed that larger sediment particles significantly alter local 3D flow structures, particularly inducing stronger vertical motions in fully developed downstream zones. This complements the theory of flow suppression by coarse sediment (> 0.5 mm) by Dorrell et al. (2018), revealing the unique feedback mechanism of fine sand under low-velocity conditions.

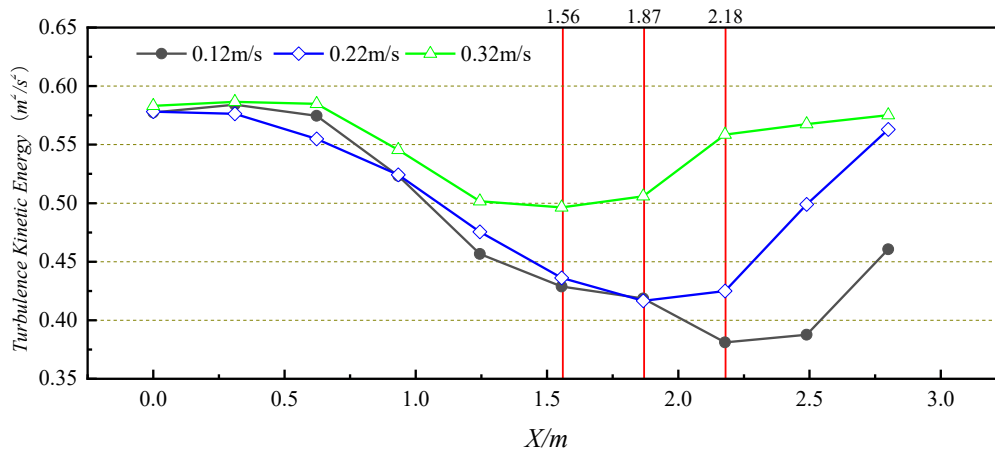


Fig. 12 TKE at different flow velocities

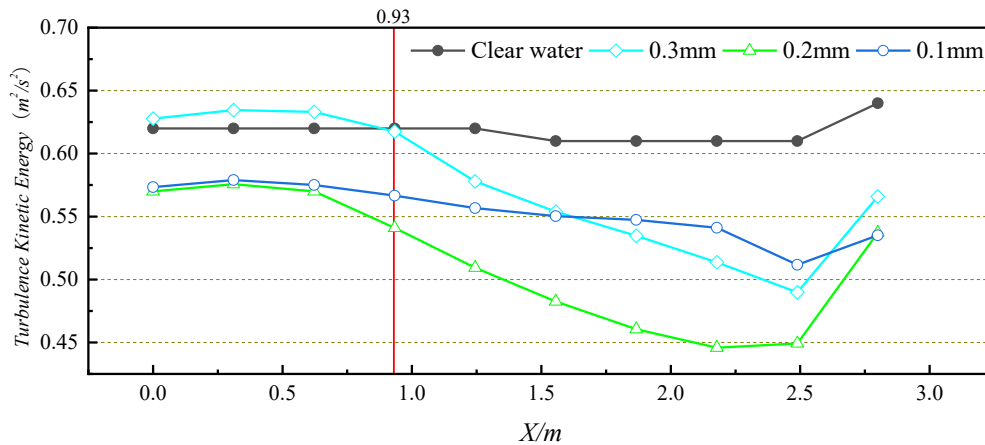


Fig. 13 Variations in TKE for different sediment particle sizes

Comprehensive analyses showed that the flow velocity and sediment particle size have a significant coupled and mutual feedback effect on the characteristics of water and sediment movement. In the coupled movement of low-flow velocity and fine particles, the flow velocity mainly controlled the intensity and direction of sediment transport (driving effect), whereas sediment particles (especially those with larger particle sizes) significantly changed the flow structure by influencing local turbulence and vertical exchange processes (feedback effect).

4.3 Mutual Feedback Mechanism of TKE for Fine Sediment and Water

TKE serves as a key variable for elucidating the physical mechanisms of water–sediment mutual feedback in low-velocity fine-sediment systems. The steady-state TKE was chosen as the baseline to exclude non-physical influences from inlet conditions and initial perturbations, ensuring statistically stable percentage comparisons. In this study, the influence of flow velocity (0.12–0.32 m/s) and sediment particle size (0.1–0.3 mm) on TKE distribution along the flume length ($X = 0.0$ – 3.0 m, inlet at $X = 3.0$ m) was systematically analysed (Figs. 12–14). Using TKE of the outlet to stabilize flow regime as the baseline, we quantified the percentage increase in TKE at other positions relative to this reference:

$$\text{TKE}_{\text{increase}, i}(\%) = \frac{\text{TKE}_i - \text{TKE}_{\text{ref}}}{\text{TKE}_{\text{ref}}} \times 100\% \quad (7)$$

where TKE_i is the TKE value at a given measurement points, and TKE_{ref} represents the TKE value under the outlet stabilized flow regime.

Figure 12 illustrates the spatial distribution of TKE along the flume ($X = 0.0$ – 3.0 m) under three flow velocities (0.12 m/s, 0.22 m/s, and 0.32 m/s) for a sediment particle size of 0.3 mm, as well as under clear-water conditions (0.32 m/s). The figure indicates that TKE remains stable under steady-state clear-water conditions, while it varies with flow velocity in the presence of sediment. In all three sediment-laden cases, TKE exhibits a non-monotonic trend, with the location of the minimum value varying with flow velocity (2.18 m at 0.12 m/s, 1.87 m at 0.22 m/s, and 1.56 m at 0.32 m/s). This suggests that at higher flow velocities (≥ 0.22 m/s), the amplitude of TKE fluctuations decreases, and TKE remains relatively high near the outlet. These observations confirm that increasing flow velocity enhances turbulence intensity, which is consistent with the linear superposition model proposed by Kim et al. (2018), indicating that higher flow velocities lead to greater TKE.

To investigate the feedback mechanism of sediment grain size on TKE, clear-water TKE was introduced for comparison (Fig. 13). The longitudinal variation characteristics of TKE along $X = 0.0$ m to $X = 3.0$ m direction were measured under three grain sizes (0.1 mm, 0.2 mm, and 0.3 mm) and clear-water conditions. The key findings indicate that peak TKE under 0.3 mm grain size

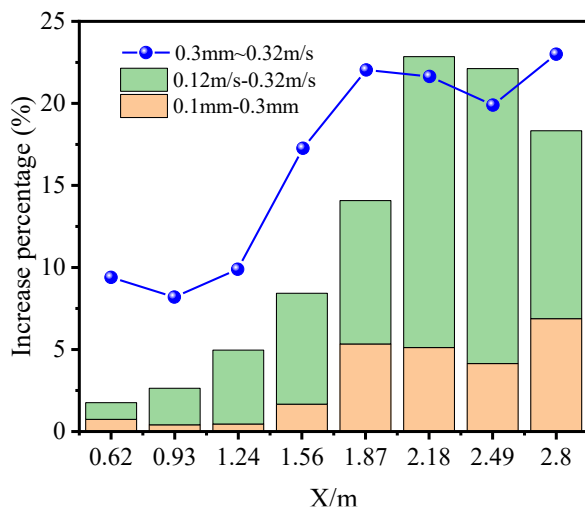


Fig. 14 Graph of TKE Increase Percentage

reached $0.68 \text{ m}^2/\text{s}^2$ near $X = 1.0 \text{ m}$; clear-water TKE was generally higher in the inlet section ($X = 3.0 \text{ m}$); under both 0.2 mm and 0.3 mm grain sizes, TKE showed a distinct low value at $X = 1.5 \text{ m}$, after which TKE for all grain sizes exhibited an initial increase followed by a decline. However, larger grain sizes ($\geq 0.2 \text{ mm}$) displayed more intense fluctuations, exceeding clear-water TKE at $X = 0.93 \text{ m}$.

The non-monotonic TKE trend is explained as follows: near the inlet ($X = 3.0 \text{ m}$), inflow jet induces high TKE; in the mid-region ($X = 2.0\text{--}2.5 \text{ m}$), turbulent dissipation dominates; downstream ($X < 2.0 \text{ m}$), sediment settling enhances density stratification, inducing secondary flow and local shear and leading to a secondary TKE rise.

Further quantifying the percentage increase in TKE (using Eq. 7) for $X = 0.0\text{--}0.3 \text{ m}$ and superimposing the results shown in Fig. 14 revealed that when the flow velocity was 0.32 m/s and particle size was simultaneously 0.3 mm , the TKE values within the deposition zone ($X < 1.87 \text{ m}$) in the flume significantly increased, with a combined average enhancement of approximately 7.23% . This amplification effect reflects the synergistic enhancement (nonlinear superposition) of TKE.

5. CONCLUSION

Through integrated numerical simulations and flume experiments, the interaction between fine-grained sediments ($0.1\text{--}0.3 \text{ mm}$) and water flow under low-velocity conditions ($0.12\text{--}0.32 \text{ m/s}$) in the upper Tarim River was elucidated. The main conclusions are as follows:

(1) The state of sediment movement is jointly controlled by particle size and flow velocity: when the sediment particle size was less than 0.2 mm and flow velocity was less than 0.12 m/s , fine sand maintained uniform transport and settling. When the particle size exceeded 0.2 mm , deposition started; when the flow velocity exceeded 0.12 m/s , the sediment deposition thickness decreased accordingly.

(2) Flow velocity primarily controls the intensity of sediment transport, whereas particle size modulates the turbulent structure: increasing the flow velocity to 0.32 m/s can increase the sediment velocity in the outlet section by 40% (reaching 0.35 m/s), whereas increasing the particle size to 0.3 mm enhanced the vertical flow velocity by 0.11 m/s . Larger particle sizes ($\geq 0.2 \text{ mm}$) had a more pronounced effect on altering the flow structure.

(3) A synergistic enhancement effect occurs between water and sediment: when both conditions, flow velocity $\geq 0.22 \text{ m/s}$ and particle size $\geq 0.2 \text{ mm}$ —are satisfied, TKE in the deposition zone of the channel increased by an additional 30% .

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CONFLICT OF INTEREST

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

AUTHOR CONTRIBUTIONS

Zihai Yang: Writing-original draft, Funding acquisition, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization; **Shunjun Hong:** Writing-review & editing, Validation, Software, Resources, Funding acquisition, Project administration, Methodology, Investigation; **Yanfu Liu:** Validation, Data curation; **Baijun Zhao:** Validation, Data curation; **Xiaozhou Hu:** Methodology, Conceptualization, Resources

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